

NUNO ANTUNES RIBEIRO

Singapore University of Technology and Design ◊ Engineering Systems and Design
08 Somapah Road 03-101 ◊ Building 1-Level 3 ◊ Singapore, 487372

ACADEMIC POSITIONS

Assistant Professor, Engineering Systems and Design Singapore University of Technology and Design (SUTD)	<i>2019–Present</i>
Deputy Director, Aviation Studies Institute Singapore University of Technology and Design (SUTD)	<i>2024–Present</i>

EDUCATION

PhD in Transportation Systems , University of Coimbra, Portugal	<i>2014–2019</i>
Visiting Scholar – Massachusetts Institute of Technology (MIT)	<i>2016</i>
Visiting Scholar – Carnegie Mellon University (CMU)	<i>2017</i>
BSc and MSc in Civil Engineering , University of Coimbra, Portugal	<i>2007–2014</i>

TEACHING EXPERIENCE

Instructor, Singapore University of Technology and Design	
10.022 Modelling Uncertainty, UG Course (Term 3) Avg. Instructor Evaluation: 4.6/5.0; Avg. Class Size: 88 students	<i>2023–2024</i>
40.015 Simulation Modelling and Analysis, UG Course (Term 6) Avg. Instructor Evaluation: 4.6/5.0; Avg. Class Size: 82 students	<i>2019–Present</i>
40.321 Airport Systems Modelling and Simulation, UG Course (Term 7) Avg. Instructor Evaluation: 4.8/5.0; Avg. Class Size: 19 students	<i>2020–Present</i>
40.614 Metaheuristic Optimization, PhD Course Avg. Instructor Evaluation: 4.9/5.0; Avg. Class Size: 19 students	<i>2022</i>
01.400 Capstone, UG Course (Term 7) Avg. Instructor Evaluation: 4.6/5.0; Total Supervised: 30 students	<i>2021</i>
01.017 SHARP Honours Session, UG Course (Term 3) Class Size: 15 students	<i>2023–2024</i>

RESEARCH GRANT SUPPORT

Lead Principal Investigator	<i>All amounts in SGD</i>
PI1. National Research Foundation Singapore, “Total Environment for Next-Gen Airport Management Experimentation Platform” Total Award: SGD 6,920,853; Lead PI for Work Package: SGD 1,158,307	<i>2026–2027</i>

- PI2. Airbus Singapore Pte., EDB IPP,
“Flight Operations Analysis and Optimization in the TMA”
Total Award: SGD 65,000 *2025–2026*
- PI3. National Research Foundation Singapore,
“Game-Based Learning in PATCO Training”
Total Award: SGD 1,104,640 *2024–2026*
- PI4. National Research Foundation Singapore,
“Evaluating CONOPS for FF-ICE R1 in South-East Asia”
Total Award: SGD 1,362,126 *2024–2026*
- PI5. National Research Foundation Singapore,
“Optimising Terminal Airspace Operations under Varying TBO Scenarios”
Total Award: SGD 1,373,438 *2024–2026*
- PI6. Civil Aviation Authority of Singapore,
“Airfield and Airspace Management”
Total Award: SGD 720,000 *2022–2024*
- PI7. National Research Foundation Singapore,
“Resilient Airspace Operations for Singapore”
Total Award: SGD 398,722 *2021–2023*
- PI8. Civil Aviation Authority of Singapore,
“Airfield Management and Economics”
Total Award: SGD 520,145 *2019–2021*
- PI9. SUTD Startup Research Grant,
“Airport Congestion Mitigation”
Total Award: SGD 100,000 *2020–2022*

Co-Principal Investigator

- CI1. National Research Foundation Singapore,
“Rapid Evaluation Tools Within the Intelligent Optimisation Platform”,
Total Award: \$472,132 *2021–2024*

International Collaboration

- IC1. External Collaborator, National Natural Science Foundation of China,
“Optimization Techniques for Collaborative and Integrated Slot Allocation for a Network of Airports”,
Total Award: \$383,943 *2025–2029*

Pre-Faculty Research Roles

- PR1. Research Assistant, European Regional Development Funds, “Airport Slot Allocation Process, Advancing Current Practices and Guidelines” *2018–2021*
- PR2. Research Assistant, European Regional Development Funds, “Energy and Mobility for Sustainable Regions, New Urban Sustainable Mobility” *2014*
- PR3. Research Consultant, Research Centre for Territory Transports and Environment (CITTA) – Valorsul S.A., “Study of transportation costs of the Valorsul Solid Waste Management System” *2013*

HONORS AND AWARDS

Research Awards

*Symbol * indicates co-authorship with supervised students*

- RA1. Winner, Best Paper Award for Future Aviation Navigation Systems (FANS) Air Transport Research Society (ATRS)* 2026
- RA2. Winner, Best Paper Award, AAAI Workshop on AI4AT* 2026
- RA3. Winner, Best Paper Award, INFORMS Air Transportation Section* 2025
- RA4. Winner, Cross-region Collaborative Grant, INFORMS Transportation Science and Logistics Society 2021
- RA5. Runner-Up, Best Paper Award, Air Transport Research Society (ATRS) 2021
- RA6. Winner, Best Paper Award, INFORMS Air Transportation Section 2020
- RA7. Winner, Best PhD Dissertation Award, INFORMS Air Transportation Section 2019
- RA8. Winner, Anna Valicek Silver Medal, AGIFORS 2018
- RA9. Runner-Up, Best Student Presentation, INFORMS Air Transportation Section 2017

Teaching Awards

- TA1. Winner, IISE Modeling and Simulation Division Teaching Award 2024

Student Awards (Supervised)

- SA1. Winner, Anna Valicek Silver Medal (Wayne Ng, PhD), AGIFORS 2025
- SA2. Winner, Best Student Presentation (Marcus Koh, PhD), INFORMS Air Transportation Section 2025
- SA3. Winner, Best Student Presentation (Wayne Ng, PhD), INFORMS Aviation Applications Section 2023

Other Honors

- OH1. Shortlisted for the CANSO ATM Awards alongside well-established aviation organisations — ENAIRE, ENAV, EUROCONTROL, ESSP, and Airways New Zealand 2023
- OH2. Competitive STaRs Program Visiting Research Grant (1-month research visit), University of Bergamo (€5,000) 2022
- OH3. PhD Scholarship, MIT-Portugal Program 2015–2019
- OH4. 3% Best Student Award, University of Coimbra 2013

PUBLICATIONS

Refereed Journal Publications

*Underlined names indicate supervised students/researchers;
Symbol * indicates corresponding author.*

- JP1. Liu, Y., Chu, N., Ng, K. H., **Ribeiro, N. A.**, Zhu, C., & Leung, C. Y. (2026). Spatio-temporal weather and airspace graph network (SWAG-Net) for trajectory-based ETA prediction. *Transportation Research Part E: Logistics and Transportation Review*, 213, 104966.
- JP2. Li, M., Birolini, S., & **Ribeiro, N. A.** (2026). Hub competition and demand modeling in long-haul aviation markets: A forward-looking scenario-based game-theoretical approach. *Transportation Research Part E: Logistics and Transportation Review*, 211, 104894.

- JP3. Yuan, M., Zhou, C., **Ribeiro, N. A.**, Ren, X., Xue, L., & Teng, Y. (2026). An exact method for storage space assignment at airport cargo terminal: A temporal-spatial packing problem. *European Journal of Operational Research*. Advance online publication.
- JP4. Li, Y., Fang, Y., Zhou, S., Long, T., Zhang, Y., **Ribeiro, N. A.**, & Melgani, F. (2025). A lightweight normalization-free architecture for object detection in high spatial resolution remote sensing imagery. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 18, 24491–24508.
- JP5. Li, Y., Fang, Y., Zhou, S., Zhang, Y., & **Ribeiro, N. A.*** (2025). Robust small-object detection in aerial surveillance via integrated multi-scale probabilistic framework. *Mathematics*, 13(14), 2303.
- JP6. Suvittawat, N., Kurniawan, C., Datephanyawat, J., Tay, J., Liu, Z., Soh, D. W., & **Ribeiro, N. A.*** (2025). Advances in aircraft skin defect detection using computer vision: A survey and comparison of YOLOv9 and RT-DETR performance. *Aerospace*, 12(4), 356.
- JP7. López-Ramos, F., Benita, F., & **Ribeiro, N. A.** (2025). A novel decision support framework for the multi-objective aircraft routing problem. *Computers & Operations Research*, 180, 107058.
- JP8. **Ribeiro, N. A.***, Tay, J., Ng, W., & Birolini, S. (2025). Delay predictive analytics for airport capacity management. *Transportation Research Part C: Emerging Technologies*, 171, 104947.
- JP9. Ng, W., **Ribeiro, N. A.***, & Jorge, D. (2024). An optimization approach for the terminal airspace scheduling problem. *Transportation Research Part C: Emerging Technologies*, 169, 104856.
- JP10. Pouget, L., **Ribeiro, N. A.**, Odoni, A. R., & Antunes, A. P. (2023). How do airlines react to slot displacements? Evidence from a major airport. *Journal of Air Transport Management*, 106, 102300.
- JP11. Birolini, S., Jacquillat, A., Schmedeman, P., & **Ribeiro, N. A.** (2023). Passenger-centric slot allocation at schedule-coordinated airports. *Transportation Science*, 57(1), 4–26.
- JP12. Jorge, D., **Ribeiro, N. A.**, & Antunes, A. P. (2021). Towards a decision-support tool for airport slot allocation: Application to Guarulhos (São Paulo, Brazil). *Journal of Air Transport Management*, 93, 102048.
- JP13. **Ribeiro, N. A.**, Jacquillat, A., & Antunes, A. P. (2019). A large-scale neighborhood search approach to airport slot allocation. *Transportation Science*, 53(6), 1772–1797.
- JP14. **Ribeiro, N. A.**, Jacquillat, A., Antunes, A. P., & Odoni, A. (2019). Improving slot allocation at Level 3 airports. *Transportation Research Part A: Policy and Practice*, 127, 32–54.
- JP15. **Ribeiro, N. A.**, Jacquillat, A., Antunes, A. P., Odoni, A., & Pita, J. (2018). An optimization approach for airport slot allocation under IATA guidelines. *Transportation Research Part B: Methodological*, 112, 132–156.

Manuscripts Under Review

- UP1. Teo, D., Jackson, P. L., Delahaye, D., & **Ribeiro, N. A.** (2025). Optimization-based simulation of air traffic flow management within the decentralized ASEAN region. *Transportation Research Part C: Emerging Technologies*. Major revision.

- UP2. Teo, D., Dai, G., Nandi, R., Jackson, P., & **Ribeiro, N. A.** (2025). From decentralized to collaborative ATFM: A simulation-based assessment of FF-ICE/R1 in Southeast Asia. *Transportation Research Part A: Policy and Practice*. Major revision.
- UP3. Ng, W., Birolini, S., & **Ribeiro, N. A.** (2026). Data-driven optimization of aircraft sequencing and descent trajectories for fuel efficiency in the terminal manoeuvring area. *European Journal of Operational Research*. Under review.
- UP4. Ng, W., Shuan, N., & **Ribeiro, N. A.** (2026). Assessing the benefits of controlled times of arrival at FIR boundaries under trajectory-based operations. *Journal of Air Transport Management*. Under review.
- UP5. **Ribeiro, N. A.**, Ng, W., & Birolini, S. (2026). Airport capacity declaration for slot allocation: A modelling framework. *Journal of Air Transport Management*. Under review.
- UP6. Yuan, Y., Zhou, C., Zhang, H., Yuan, M., Lee, B. K., **Ribeiro, N. A.**, & Teng, Y. (2026). Online sorting and packing with recirculation: A threshold-adaptive framework for multi-robot baggage handling systems. *European Journal of Operational Research*. Under review.

Peer-Reviewed Conference Publications

- CP1. Teo, D., Sam, A., Koh, C. S. M., Nagi, R., & **Ribeiro, N. A.** (2026). Semi-automated knowledge engineering and process mapping for total airport management. In *Proceedings of the 29th International Conference on Information Fusion (FUSION 2026)*. IEEE.
- CP2. Jackson, P., & **Ribeiro, N. A.** (2024). Terminal maneuvering area design tool: Translational research in aviation. In *Proceedings of the 5th Asia Pacific Conference on Industrial Engineering and Operations Management*.
- CP3. Suvittawat, N., & **Ribeiro, N. A.** (2024). Aircraft surface defect inspection system using AI with UAVs. In *Proceedings of the International Conference on Research in Air Transportation (ICRAT)*.
- CP4. Ng, W., & **Ribeiro, N. A.** (2024). A modeling framework for rapid assessment of terminal airspace configurations and planning scenarios. In *Proceedings of the International Conference on Research in Air Transportation (ICRAT)*.
- CP5. Tralhão, L., Sousa, N., **Ribeiro, N. A.**, & Coutinho-Rodrigues, J. (2014). Design of bicycling suitability maps for hilly cities. *Proceedings of the Institution of Civil Engineers – Municipal Engineer*, 168(2), 96–105.
- CP6. **Ribeiro, N. A.**, Tralhão, L., Coutinho-Rodrigues, J., & Sousa, N. (2013). Building a bicycle suitability map for Coimbra. In *Proceedings of the Energy for Sustainability Multidisciplinary Conference*.

Technical Reports and White Papers

- TP1. Alzurikat, M., & Chng, S. (2024). *A future with autonomous urban air mobility* [ASI White Paper]. Singapore University of Technology and Design. Commissioning editors: J. Bloomfield & **N. A. Ribeiro**.
- TP2. Sobie Aviation. (2024). *Airport–rail inter-modality in Southeast Asia: Opportunities and the impact of high-speed rail* [ASI White Paper]. Singapore University of Technology and Design. Commissioning editors: J. Bloomfield & **N. A. Ribeiro**.

- TP3. Sobie Aviation. (2024). *The future of super short-haul air travel in Southeast Asia* [ASI White Paper]. Singapore University of Technology and Design. Commissioning editors: J. Bloomfield & **N. A. Ribeiro**.
- TP4. International Air Transport Association. (2022). *Slot allocation amidst COVID-19 recovery* [ASI White Paper]. Commissioning editors: J. Bloomfield & **N. A. Ribeiro**.

TALKS AND PRESENTATIONS

Invited Lectures and Seminars

- IT1. Guest Lecturer, Airport Planning Programme, Singapore Aviation Academy (CAAS), Singapore, Three sessions on “Airside Capacity” 2026
- IT2. Panellist, Roundtable, Super Terminal Expo, Hong Kong, “Rethinking Apron Efficiency for Faster Turnarounds” 2025
- IT3. Guest Speaker, Singapore Aviation Research and Technology Exchange Day (SAR-TED), “From Airspace to Apron: Informing Decisions Through Analytics” 2025
- IT4. Guest Speaker, Seminar Series, School of Economics and Management, Tongji University, China, “From Predictive to Prescriptive Analytics in Airport Capacity Management” 2025
- IT5. Guest Speaker, Seminar Series, College of Civil Aviation, Nanjing University of Aeronautics and Astronautics, China, “From Predictive to Prescriptive Analytics in Airport Capacity Management” 2025
- IT6. Guest Speaker, Seminar Series, School of Management, Northwestern Polytechnical University, China, “From Predictive to Prescriptive Analytics in Airport Capacity Management” 2025
- IT7. Guest Speaker, ASI Symposium, Aviation Trends and Developments in Air Traffic Management, “Resilient Airspace Research” 2025
- IT8. Guest Speaker, Seminar Series, Department of Aeronautical and Aviation Engineering, The Hong Kong Polytechnic University, Hong Kong, “From Predictive to Prescriptive Analytics in Airport Capacity Management” 2024
- IT9. Guest Speaker, European Union Aviation Safety Agency (EASA) Artificial Intelligence Workshop for the Asian Region, Singapore, “Artificial Intelligence Applications in Aviation Analytics” 2024
- IT10. Guest Lecturer, Air Transport Management Degree Programme (ATM3307 Operations Research), Singapore Institute of Technology, Singapore, “Optimization and Analytics in Air Transport Operations” 2024
- IT11. Guest Speaker, Seminar Series, School of Data Science, City University of Hong Kong, Hong Kong, “From Predictive to Prescriptive Analytics in Airport Capacity Management” 2024
- IT12. Guest Speaker, Seminar Series, Air Transport and Operations Group, Faculty of Aerospace Engineering, Delft University of Technology, Delft, Netherlands, “Passenger-Centric Slot Allocation and Airport Capacity Analytics” 2023
- IT13. Guest Speaker, GARS Research Seminar, Amsterdam University of Applied Sciences, Amsterdam, Netherlands, “How to Make Slot Allocation Work Better?” 2023
- IT14. Guest Speaker, Seminar Series, Department of Management, Information and Production Engineering, University of Bergamo, Bergamo, Italy, “Passenger-Centric Slot Allocation and Airport Capacity Management” 2022
- IT15. Guest Speaker, Seminar Series, Department of Logistics and Maritime Studies, The Hong Kong Polytechnic University, Hong Kong / Virtual, “Passenger-Centric Slot Allocation at Schedule-Coordinated Airports” 2021
- IT16. Invited Speaker, Civil Air Navigation Services Organisation (CANSO) Webinar, ATM Research and Development Series, Online, “Understanding Convective Weather Events and Their Impact on Air Traffic Management” 2020

Conference Presentations

- CT1. 29th Air Transport Research Society (ATRS) World Conference, “Data-driven Optimization of Aircraft Sequencing and Descent Trajectories for Fuel Efficiency in the Terminal Manoeuvring Area”, Beijing, China 2026
- CT2. 2025 INFORMS Annual Meeting, “An Optimization Approach for the Terminal Airspace Scheduling Problem,” Atlanta, USA 2025
- CT3. 2025 INFORMS Annual Meeting, “Prescriptive Analytics for Optimizing Airport Slot Allocation,” Atlanta, USA 2025
- CT4. 28th Air Transport Research Society (ATRS) World Conference, “Modeling Demand Competition in Long-Haul Aviation Markets,” Hong Kong 2025
- CT5. 2025 INFORMS International Meeting, “Airport Capacity Declaration for Slot Allocation: A Modelling Framework,” Singapore 2025
- CT6. 2024 INFORMS Annual Meeting, “Modeling Demand Competition in Long-Haul Aviation Markets,” Seattle, USA 2024
- CT7. 5th Asia Pacific Conference on Industrial Engineering and Operations Management (IEOM), “Terminal Maneuvering Area Design Tool: Translational Research in Aviation,” Tokyo, Japan 2024
- CT8. 11th International Conference on Research in Air Transportation (ICRAT), “A Modeling Framework for Rapid Assessment of Terminal Airspace Configurations and Planning Scenarios,” Singapore 2024
- CT9. 26th Air Transport Research Society (ATRS) World Conference, “Hub Competition and Demand Modeling in Long-Haul Aviation Markets,” Kobe, Japan 2023
- CT10. 2022 INFORMS Annual Meeting, “A Modelling Approach to Optimize Slot Limits in Airport Demand Management,” Indianapolis, USA 2022
- CT11. 2021 INFORMS Annual Meeting, “Optimizing the Declared Capacity at Slot Coordinated Airports,” Online 2021
- CT12. 22nd Conference of the International Federation of Operational Research Societies (IFORS), “A Passenger-Centric Approach to Airport Slot Allocation,” Online 2021
- CT13. 25th Air Transport Research Society (ATRS) World Conference, “Passenger-Centric Slot Allocation at Schedule-Coordinated Airports,” Online 2021
- CT14. IORA International Virtual Conference on Operations Research 2021, “Predictive and Prescriptive Analytics for Airport Slot Allocation,” Online 2021
- CT15. 31st European Conference on Operational Research (EURO), “Passenger-Centric Slot Allocation at Schedule-Coordinated Airports,” Online 2021
- CT16. 2020 INFORMS Annual Meeting, “A Passenger-Centric Approach to Airport Slot Allocation,” Online 2020
- CT17. 2019 INFORMS Annual Meeting, “Slot Allocation: Algorithms and Implementation at Guarulhos Airport,” Seattle, USA 2019
- CT18. 2018 INFORMS Annual Meeting, “A Large Neighborhood Search Approach to Airport Slot Allocation,” Phoenix, USA 2018
- CT19. 58th Annual Symposium of the Airline Group of the International Federation of Operational Research Societies (AGIFORS), “An Optimization Approach for Airport Slot Allocation Under IATA Guidelines,” Tokyo, Japan 2018
- CT20. 60th Annual Conference of the Operational Research Society (OR60), “Optimizing Slot Allocation at Level 3 Airports,” Lancaster, UK 2018
- CT21. 2017 INFORMS Annual Meeting, “An Optimization Approach for Airport Slot Allocation Under IATA Guidelines,” Houston, USA 2017
- CT22. 2016 INFORMS Annual Meeting, “A Model for Airport Schedule Coordination Based on the IATA Guidelines,” Nashville, USA 2016

- CT23. 5th Symposium of the European Association for Research in Transportation (hEART), “An Airport Slot Allocation Model Compliant with IATA Guidelines,” Delft, Netherlands 2016

Student Conference Presentations

- ST1. 29th Air Transport Research Society (ATRS) World Conference, “A Vehicle Routing and Scheduling Approach to Airside Baggage Handling Operations”, Marcus Koh (PhD), Beijing, China 2026
- ST2. The 40th Annual AAAI Conference on Artificial Intelligence – Workshop on Artificial Intelligence for Air Transportation (AI4AT), Wayne Ng (Research Fellow), “Data-Driven Terminal Airspace Optimization with Environmental Considerations,” Singapore 2026
- ST3. 65th Annual Symposium of the Airline Group of the International Federation of Operational Research Societies (AGIFORS), Wayne Ng (PhD), “An Optimization Approach for the Terminal Airspace Scheduling Problem,” London, UK 2025
- ST4. 2025 INFORMS Annual Meeting, Marcus Koh (PhD), “Towards Efficient Aircraft Turnaround: A Mesoscopic Model for Equipment Routing and Scheduling,” Atlanta, USA 2025
- ST5. INFORMS Annual Meeting, Marcus Koh (PhD), “Optimizing Baggage and Cargo Transport for Aircraft Turnaround Process,” Atlanta, USA 2025
- ST6. Singapore Analytics & Operations Day, Wayne Ng (PhD), “Optimizing Aircraft Flows and Trajectories in the Terminal Airspace with Environmental Considerations,” Singapore 2025
- ST7. 2025 INFORMS International Meeting, Darryl Teo (PhD), “Simulated Impact of Information Sharing and Collaboration Within the Decentralized ASEAN Region,” Singapore 2025
- ST8. 2025 INFORMS International Meeting, Jetanat Datephanyawat (PhD), “Optimizing Aircraft Trajectories in Terminal Airspace Under Convective Weather Conditions,” Singapore 2025
- ST9. 2025 INFORMS International Meeting, Marcus Koh (PhD), “Optimizing System-Level Aircraft Turnaround Operations,” Singapore 2025
- ST10. 2025 INFORMS International Meeting, Wayne Ng (PhD), “Optimizing Aircraft Flows and Trajectories in the Terminal Airspace with Environmental Considerations,” Singapore 2025
- ST11. 2025 INFORMS International Meeting, Mingmei Li (Research Fellow), “Hub Competition and Demand Modeling in Long-Haul Aviation Markets,” Singapore 2025
- ST12. 28th Air Transport Research Society (ATRS) World Conference, Wayne Ng (PhD), “Optimizing Aircraft Flows and Trajectories in the Terminal Airspace with Environmental Considerations,” Hong Kong 2025
- ST13. 28th Air Transport Research Society (ATRS) World Conference, Darryl Teo (PhD), “Optimizing Aircraft Flows and Trajectories in the Terminal Airspace with Environmental Considerations,” Hong Kong 2025
- ST14. 13th OpenSky Symposium, Wayne Ng (PhD), “Data-Driven Optimization of Aircraft Sequencing and Descent Trajectories for Fuel Efficiency in the Terminal Manoeuvring Area,” Norrköping, Sweden 2025
- ST15. 2024 INFORMS Annual Meeting, Wayne Ng (PhD), “Optimizing Aircraft Flows and Fuel Savings in Terminal Airspace Operations,” Seattle, USA 2024
- ST16. 2024 INFORMS Annual Meeting, Darryl Teo (PhD), “Optimization-Based Simulation of Target Time Over For Flights Within the Decentralized ASEAN Region,” Seattle, USA 2024

- ST17. 2024 INFORMS Annual Meeting, Marcus Koh (PhD), “Optimizing System’s Level Aircraft Turnaround Operations,” Seattle, USA 2024
- ST18. 2024 INFORMS Annual Meeting, Jetanat Datephanyawat (PhD), “Terminal Airspace Operations under Convective Weather Uncertainty,” Seattle, USA 2024
- ST19. 27th Air Transport Research Society (ATRS) World Conference, Wayne Ng (PhD), “An Optimization Approach for Solving the Terminal Airspace Scheduling Problem,” Lisbon, Portugal 2024
- ST20. 5th Symposium on Aviation Research (SoAR), Wayne Ng (PhD), “Terminal Airspace Scheduling Optimization with Environmental Considerations,” Luxembourg 2024
- ST21. 2023 INFORMS Annual Meeting, Wayne Ng (PhD), “The Terminal Airspace Sequencing and Scheduling Problem: An Optimization Approach,” Phoenix, USA 2023
- ST22. 26th Air Transport Research Society (ATRS) World Conference, Wayne Ng (PhD), “Strategic Optimization Model for Resilient Terminal Airspaces,” Kobe, Japan 2023
- ST23. 2022 INFORMS Annual Meeting, Wayne Ng (PhD), “Strategic Model to Optimize Terminal Airspace Operations,” Indianapolis, USA 2022
- ST24. 25th Air Transport Research Society (ATRS) World Conference, Wayne Ng (PhD), “Optimizing Air Traffic Operations in Congested Terminal Manoeuvring Areas,” Virtual 2022

STUDENT SUPERVISION

Doctoral Students

1. **Marcus Koh**
Towards Efficient Airport Ground Operations: Models for Resource Allocation and Scheduling
Ongoing. 2023–Present
2. **Jetanat Datephanyawat**
Terminal Airspace Operations Under Convective Weather Uncertainty
Ongoing. 2022–Present
3. **Wayne Ng**
Towards an Optimization Framework for Efficient and Sustainable Terminal Airspace and Airfield Operations
Graduated. Placement after graduation: Research Fellow, SUTD 2021–2025
4. **Darryl Teo**
Quantifying the Value of Information and Collaboration in the Decentralized ASEAN Airspace Network
Graduated. Placement after graduation: Research Fellow, SUTD 2020–2025

Visiting Doctoral Students

1. **Yuan Mengxue**
Research on Space Allocation and Task Scheduling in Air Cargo Terminals
Xi’an Polytechnic University, Visited SUTD from September 2024 to March 2025
Ongoing. 2022–Present
2. **Lorenzo Zanchi**
Mesoscopic Holistic Approach to Airport Terminal Modelling
Bergamo University, To visit SUTD from September 2026 to December 2026
Ongoing. 2025–Present

Master's Students

1. **Nicole Shuan**
Algorithmic Approaches for Rapid Optimization of Airport Ground Operations
Ongoing. 2026–Present
2. **Jie Xun**
Flight Operations Analysis and Optimization in the Terminal Maneuvering Area (TMA)
Ongoing. IDB-EPP Program, Airbus 2025–Present
3. **Jordan Tay**
From Prediction to Optimization: Analytics for Airport Management
Ongoing. 2023–Present
4. **Xie Peining**
Optimizing the Setting of Declared Capacities at Slot Coordinated Airports
Graduated. First Position: Engineer, Scoot Pte Ltd 2021–2022

Supervision of Student Projects

1. **UROP Project:** *Video Analytics for Aircraft Turnaround Operations Using Deep Learning*
Students: Justin Chong Zhe Wei, Tay Sai Kok 2026
2. **UROP Project:** *Optimisation of Airport Ground Operations*
Students: Kanav Dhingra, Josiah Phua Kai-Xuan 2026
3. **UROP Project:** *Semi-Automated Knowledge Engineering and Process Mapping for Total Airport Management*
Students: Justin Chong Zhe Wei, Tay Sai Kok 2026
4. **SHARP Honours Project (Terms 6–8):** *Enhancing Aircraft Turnaround Efficiency Through Data Analytics and Process Optimization*
Student: Jonathan Ong 2026
5. **MTD Data Science Design Project:** *Predicting Flight Durations and Trajectories*
Students: Sun Yongbo; Aryutkin Alexandr; Nuraini Zailani 2025
6. **SHARP Honours Project (Terms 4–5):** *Optimizing Aircraft Turnaround Processes*
Student: Jonathan Ong 2025
7. **UROP Project:** *Investigating the Effects of New Wake Turbulence Separation Concepts for Improved Airport Operations*
Students: Melvin Chong; Andy Low Zhi Hao; Teo Yu Wen
Award: LCC Poster Award 2023
8. **SHARP Honours Project (Term 3):** *Modelling Flight Delays Across Southeast Asia*
Students: Mike Umali; Xie Jiacheng 2022
9. **UROP Project:** *Airport Congestion Delays in the ASEAN Region*
Students: Cheong Cher Lynn; Lim Aziz Bin Adam; Subesh Kumar; Muhammad Zulfiqar; Ong Kai Xin Chloe; Su Chang Yi Tang
Award: LCC Poster Award 2021
10. **Internship (ASI):** *Convective Weather Events and Air Traffic Management*
Student: Richard Huang 2021

11. **SHARP Honours Project (Term 3):** *Modeling Aircraft Runway Movements at Changi Airport*
Students: Long Tran; Liu Renhang 2021
12. **UROP Project:** *Airport Capacity Study: Application to ASEAN Airports*
Students: Ryan Pan Tang Kai; Joseph Lai Lok Hong; Chen Jiasen; Ong Tao Jie; Jyotit Kaushal; Dingal Lim Jing Li; Yang Yinyu
Awards: LCC Poster Award; Finalist, SGTech Techblazer Awards 2020
13. **Internship (ASI):** *Airport Capacity Study: Application to ASEAN Airports*
Student: Mark Feng Zhengqing 2020
14. **UROP Project:** *Building an Airside Simulation Model for Changi Airport*
Students: Shyam Sridhar; Lim Ze Li 2020
15. **SHARP Honours Project (Term 6):** *Optimizing Airport Capacity Utilization*
Student: Vanessa Chia Yun Yao 2020

POSTDOCTORAL AND RESEARCH STAFF SUPERVISION

Current Research Team

1. **Wayne Ng**
Research Fellow, Projects Involved: PI1; PI5 2025–Present
2. **Darryl Teo**
Research Fellow, Projects Involved: PI1; PI4 2025–Present
3. **Mingmei Li**
Research Fellow, Projects Involved: PI5; PI6 2022–Present
4. **Marcus Koh**
Senior Research Assistant, Projects Involved: PI1; PI3 2024–Present
5. **Weimeng Wang**
Senior Research Assistant, Projects Involved: PI3 2024–Present
6. **Amul Kumar**
Senior Research Assistant, Projects Involved: PI3 2024–Present
7. **Jordan Tay**
Senior Research Assistant, Projects Involved: PI3; PI8 2021–Present
8. **Ge Nee**
Research Officer, Projects Involved: PI1 2025–Present
9. **Adharsha Sam**
Research Officer, Projects Involved: PI1 2025–Present
10. **Nicole Shuan**
Research Officer, Projects Involved: PI1; PI5 2024–Present
11. **Sian Poh**
Research Manager, Projects Involved: PI3 2024–Present

Former Research Staff

1. **Quang Thinh Dinh**
Research Fellow
Next appointment: Optimization Modeling Lead, Vinsmart Future 2023–2025

2. **Rakesh Nandi**
Research Fellow
Next appointment: Independent Data Analytics Consultant 2024–2025
3. **Aasheesh Dixit**
Research Fellow
Next appointment: Senior Scientist, Infor 2022–2023
4. **Karkulali Pugalenth**
Research Fellow
Next appointment: Scientist, A*STAR SimTech 2022–2023
5. **Padmapriya Srinivasan**
Senior Research Assistant 2024–2025
6. **Dai Gengling**
Senior Research Assistant
Next appointment: Independent Data Analytics Consultant 2024–2025
7. **Wenhe Cao**
Research Officer
Next appointment: Flight Operations Coordinator, All Nippon Airways 2022
8. **Yang Yinyu**
Research Officer
Next appointment: Co-Founder, Rosen Bridge Singapore Pte Ltd 2021–2024
9. **Siew Ning**
Research Officer
Next appointment: Systems Engineer, MITRE 2024–2025
10. **Wan Ying**
Research Officer 2020–2021

GRADUATE COMMITTEE SERVICE

Doctoral Thesis Committees

1. **Ng Jia Yi**
Robust Optimal Design of Urban Drainage Systems
Thesis Advisor: Associate Professor Stefano Galelli 2020
2. **Vu Trung Dung**
Using Satellite Observations to Address Power Asymmetries in Transboundary River Basins
Thesis Advisor: Associate Professor Stefano Galelli 2023
3. **Meijing Zhang**
Investigating the Integration of Freight and Public Transport
Thesis Advisor: Associate Professor Lynette Cheah 2023
4. **Christian Kurniawan**
Machine Paradigms in Action: Microfinance Decision Making Optimization, Electric Vehicle Battery Swapping Assignment, and Computer Vision Tuning
Thesis Advisor: Assistant Professor Soh De Wen
Status: Expected 2027

Master's Thesis Committees

1. **Pei Jinling**
Visualization of the Impact of Convective Weather Events on Air Traffic Rerouting in Southeast Asia
Thesis Advisor: Professor Peter Jackson 2021
2. **Zeng Xinyu**
Arts Places in Singapore and Their Socio-Economic Benefits
Thesis Advisor: Associate Professor Bige Tuncer 2021
3. **Teresa Liew Si Min**
Technology Acceptance and Commercialisation of Computerized Cognitive Intervention Tools Among Older Adults
Thesis Advisor: Associate Professor Yow Wei Quin 2022
4. **Sean Koh**
In-Store vs. Home Delivery Routing in Offline-to-Online Retail Delivery
Thesis Advisor: Assistant Professor Xiaotang Yang
Status: Expected 2027

UNIVERSITY SERVICE

1. **Deputy Director, Aviation Studies Institute (ASI)** 2024–Present
Principal Investigator, Aviation Studies Institute (ASI) Since 2019
 - Oversight of institute operations, research planning, and programme coordination.
 - Contributed to the growth of ASI's research portfolio (SGD 6,641,379 secured).
 - Average research team size under supervision: 10–12 researchers.
 - Quarterly reporting to the Governing Council and Advisory Board.
 - Led ASI outreach and external engagement initiatives.
 - Coordinated programme office activities and institute-wide initiatives.
 - Supported the organization of the ASI Symposium, SAR-TED, and related events.
2. **Lead, Aviation Systems Specialization Track** 2019–Present
 - Developed undergraduate course 40.321 *Airport Systems Modelling and Simulation*.
 - Conducted advising and information sessions for students.
 - Served as primary point of contact for the specialization track.
3. **Lead, ESD Research and Scholarly Excellence (RASE)** 2025–Present
Member, ESD Research and Scholarly Excellence (RASE) Since 2023
 - Coordinated research initiatives within the ESD pillar.
 - Improved visibility of research-related information on the department website.
 - Identified and engaged potential collaborators and external stakeholders.
 - Contributed to mid- and long-term research planning.
4. **Curriculum Contributions to ESD** 2019–Present
 - Designed undergraduate course 40.321 *Airport Systems Modelling and Simulation*.
 - Designed graduate course 40.614 *Metaheuristic Optimization*.
 - Contributed to the design of course 40.018 *Heuristics and Systems Theory*.
5. **Main Instructor, SHARP Honours Session** 2023–2024
 - Designed course 01.017 *SHARP Honours Session on Metaheuristic Optimization*.
6. **Club Advisor, SUTD Aviation Interest Group** 2024–Present

7. **University and Department Outreach Activities** *2019–Present*
 - Contributed to major SUTD events, including Open House, admissions activities, receptions, career talks, the Singapore Airshow, and outreach to junior college students.
 - Participated in outreach videos for Open House 2020 and SUTD Explain.
 - Supported development of the game *ApronFlow.opt* for SUTD Open House 2026.
8. **Undergraduate Admissions Information Sessions** *2020–Present*
9. **Transfer Credit Approvals for Exchange Students** *2020–Present*
10. **Member, Capstone Project Scoping Committee** *2020–2021*
11. **Member, ESD PhD Qualifying Exam Committee (Linear Optimization)** *2021*

PROFESSIONAL SERVICE

1. **Leadership, Air Transportation Section (AAS), INFORMS**
 - Vice-President (Elected) *2024–Present*
 - Treasurer / Secretary (Elected) *2023–2024*
 - Communication Chair *2021–2023*
2. **Editorial Board Membership**
 - *Transportation Research Part C: Emerging Technologies* *2021–Present*
 - *Journal of Air Transport Management* *2025–Present*
3. **Conference and Workshop Organization**
 - Co-Organizer, AAAI Workshop on AI4AT (7 Talks, 16 Posters) *2020*
 - Cluster Chair, INFORMS Annual Meeting (11 Sessions, 42 talks) *2020*
 - Cluster Chair, INFORMS International Meeting (11 sessions, 42 talks) *2025*
 - Session Chair, INFORMS Annual Meeting (4 Talks each) *2019, 2020, 2021*
 - Session Chair, IFORS World Congress (4 Talks) *2021*
4. **Award Committees**
 - Committee Chair, Best Paper Award, ATS, INFORMS *2020–2021*
 - Member, Evaluation Committee, Anna Valicek Award, AGIFORS *2021*
 - Member, Evaluation Committee, Ken Wang Award, AGIFORS *2021*
5. **Journal Reviewer**
 - *Operations Research*;
 - *European Journal of Operational Research*;
 - *Transportation Research Part B*;
 - *Transportation Research Part C*;
 - *Transportation Research Part D*;
 - *Transportation Research Part E*;
 - *Transportation Science*;
 - *Journal of Air Transport Management*;
 - *Journal of Aerospace Technology and Management*;
 - *Mathematical Problems in Engineering*

6. Professional Memberships

- INFORMS (Transportation Science and Logistics; Air Transportation Section)
- Institute of Industrial and Systems Engineers (IISE)
- Civil Air Navigation Services Organisation (CANSO)
- Singapore Institute of Aerospace Engineers (SIAE)
- Research Centre for Territory, Transports and Environment (CITTA)